

Cut Locus of Submanifolds: Stability and Intersection with Focal Locus

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Presented at

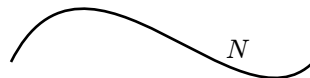
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Introduction

Given a submanifold N of a Riemannian manifold (M, g) , and a point $q \in M$, an N -segment joining N to q is a unit-speed geodesic $\gamma : [0, L] \rightarrow M$, such that

$$p := \gamma(0) \in N, \quad \gamma(L) = q, \quad \text{Len}(\gamma) = L = \text{dist}_g(N, q).$$



Definition

The **cut locus** $\text{Cut}(N)$ of a submanifold $N \subset M$ is the set of points $q \in M$ such that there exists an N -segment joining N to q , any extension of which fails to be distance-minimizing from N .

$$M \setminus \text{Cut}(N) \underset{\text{s.d.r}}{\simeq}, \quad M \setminus N \underset{\text{s.d.r}}{\simeq} \text{Cut}(N).$$

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Theorem : (Prasad [2024])[†]

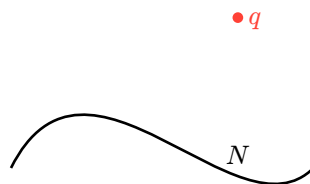
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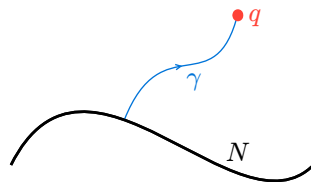
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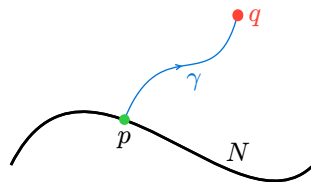
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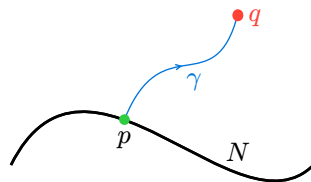
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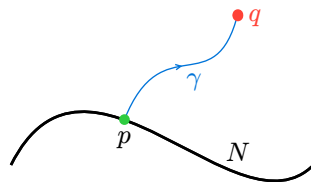
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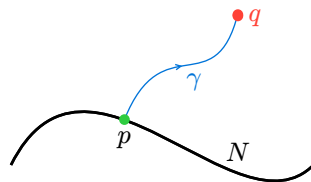
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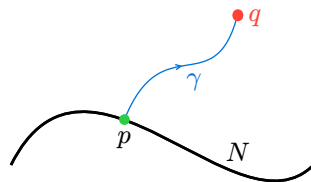
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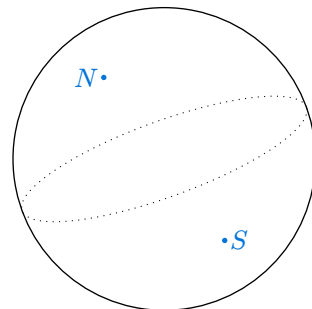
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Examples

The cut locus of the south pole S on the sphere S^2 consists of the north pole N .

More generally, the cut locus of a point $p \in S^n$ is the singleton $\{-p\}$



The cut locus of the equatorial S^1 in the sphere S^2 is the set $\{N, S\}$, which is S^0 .

More generally, the cut locus of $S^k \subset S^n$ is S^{n-k-1}

The cut locus of $\mathbb{RP}^k \subset \mathbb{RP}^n$ is $\mathbb{RP}^{n-k-1}$, and the cut locus of $\mathbb{CP}^k \subset \mathbb{CP}^n$ is $\mathbb{CP}^{n-k-1}$.

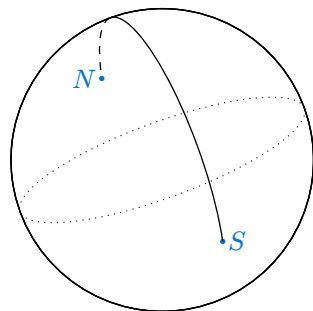
The cut locus of $O(n, \mathbb{R})$ in $M(n, \mathbb{R})$ is $M(n, \mathbb{R}) \setminus \text{GL}(n, \mathbb{R})$, the set of singular matrices (Basu-Prasad [2023][†]). This gives a geometric interpretation of the Gram-Schmidt process.

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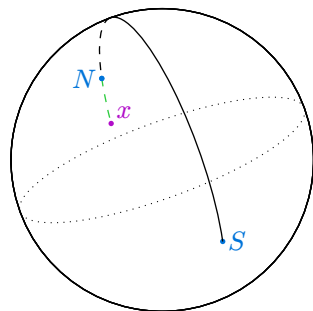
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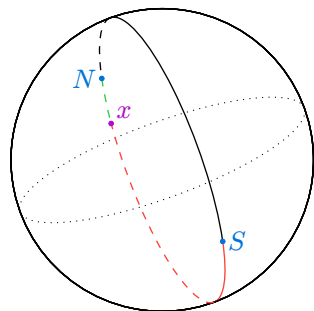
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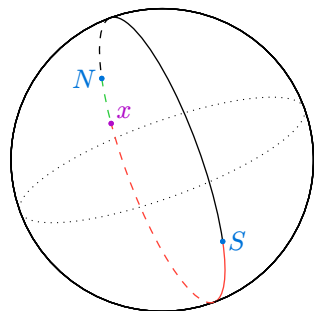
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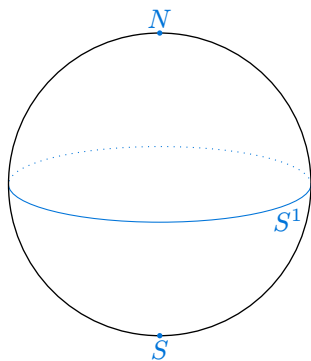
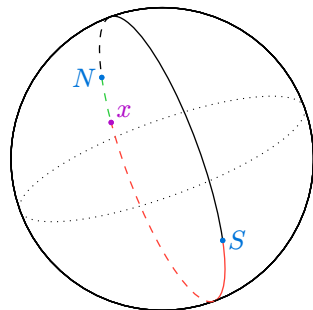
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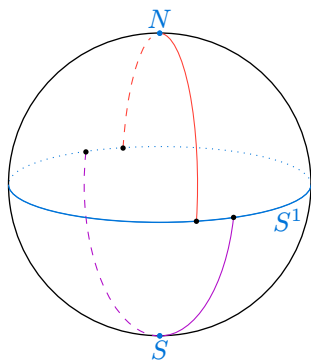
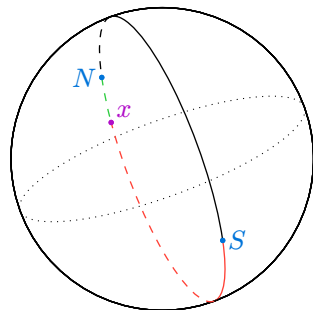
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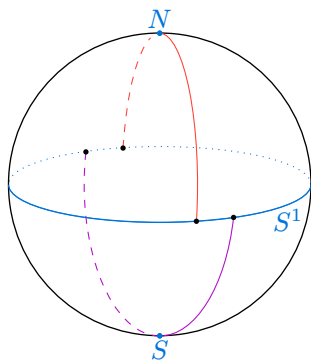
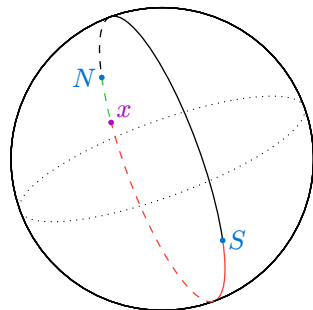
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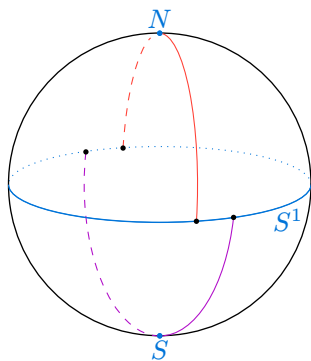
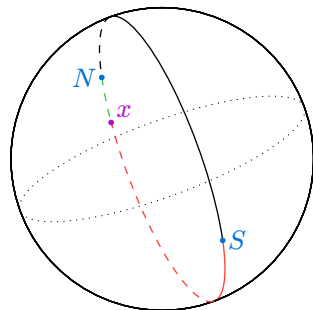
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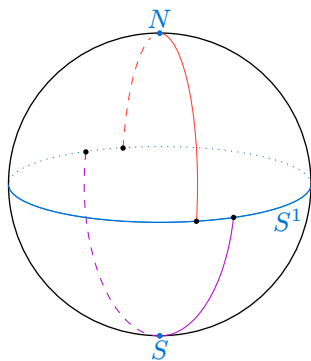
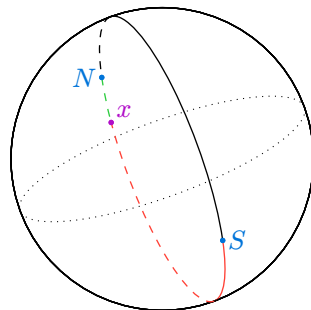
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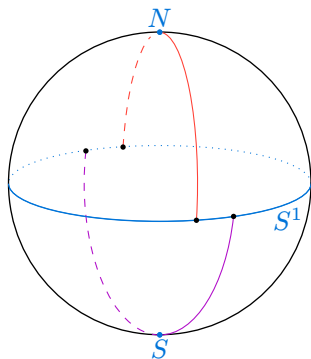
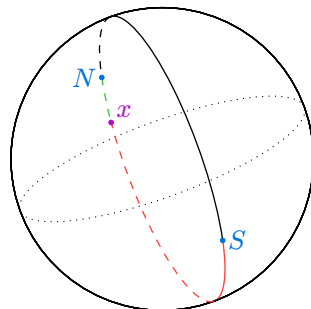
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[†] Somnath Basu and Sachchidanand Prasad. *A connection between cut locus, Thom space and Morse-Bott functions*. *Algebr. Geom. Topol.* (2023).

Cut Time and Injectivity Radius

Given a unit normal vector $\mathbf{n} \in TN^\perp_{\mathfrak{g}}$, the **cut time** is defined as

$$\rho_{\mathfrak{g}}(N, \mathbf{n}) := \sup\{t \mid \text{dist}_{\mathfrak{g}}(N, \gamma_{\mathbf{n}}^{\mathfrak{g}}(t)) = t\},$$

where $\gamma_{\mathbf{n}}^{\mathfrak{g}}$ is the unique geodesic w.r.t. $\mathfrak{g}$, with initial velocity $\mathbf{n}$.

Given $N \subset M$, the **injectivity radius** of N is defined as

$$\text{Inj}(N, \mathfrak{g}) := \text{dist}_{\mathfrak{g}}(N, \text{Cut}(N, \mathfrak{g})).$$

The injectivity radius of $(M, \mathfrak{g})$ is defined as

$$\text{Inj}(\mathfrak{g}) := \inf_{p \in M} \text{Inj}(p, \mathfrak{g}).$$

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Theorem : (Sakai [1983][†])

Suppose $\mathfrak{g}_i$ is a sequence of Riemannian metrics on a closed manifold M , converging to the metric $\mathfrak{g}_0$ in the Whitney C^2 topology. Then, $\lim_i \text{Inj}(\mathfrak{g}_i) = \text{Inj}(\mathfrak{g}_0)$. In other words, $\mathfrak{g} \mapsto \text{Inj}(\mathfrak{g})$ is continuous.

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Stability of the Cut Locus

The stability of the cut locus is well-studied in the literature from different perspectives.

- **Buchner (1977):** $\text{Cut}(p, g)$ is stable from the point of view of singularity theory for generic g , on a manifold of dimension ≤ 6 .
- **Arday-Guijarro (2011):** For an open set $\Omega \subset M$ with boundary $\partial\Omega$, $\text{Sep}(\partial\Omega, g) \cap \Omega$ is Hausdorff stable when the boundary $\partial\Omega$ is perturbed by C^2 -small diffeomorphisms.
- **Albano (2016):** $\text{Cut}(\partial\Omega, g)$ is Hausdorff stable, when both the boundary $\partial\Omega$ of a bounded domain $\Omega \subset \mathbb{R}^n$, and the metric are perturbed in a C^2 -small sense.
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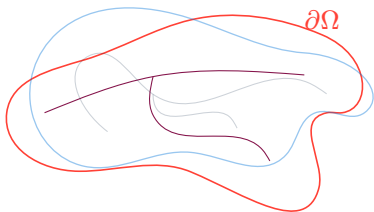
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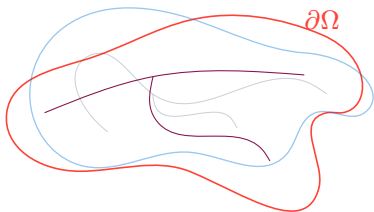
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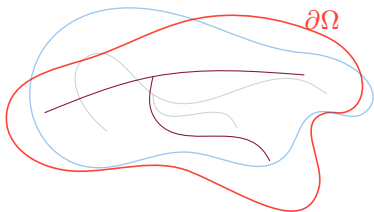
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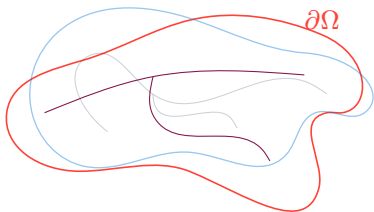
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Cut Locus as a Map

Given closed sets $A, B \subset M$ of a metric space (M, d) , the **Hausdorff distance** between them is

$$d_H(A, B) := \max \left\{ \sup_{a \in A} \text{dist}(a, B), \sup_{b \in B} \text{dist}(A, b) \right\}.$$

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- Given a manifold N , let $\text{Emb}(N, M)$ be the space of smooth embeddings $N \hookrightarrow M$, equipped with the Whitney C^2 -topology.

As the cut locus is closed, we have the maps

$$\begin{aligned} \text{Cut} : M \times \mathcal{G} &\rightarrow \mathcal{K} \\ (p, \mathfrak{g}) &\mapsto \text{Cut}(p, \mathfrak{g}), \end{aligned}$$

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Let M be a compact manifold without boundary, and $\mathfrak{h}$ be an auxiliary metric on M .

- Let $\mathcal{G} = \mathcal{G}(M)$ be the collection of C^2 Riemannian metrics on M , equipped with the Whitney C^2 -topology.
- Let $\mathcal{K} = \mathcal{K}(M) := \{K \subset M \mid K \text{ is compact}\}$, equipped with the topology induced by the Hausdorff metric d_H associated with $\mathfrak{h}$. The topology is independent of the choice of $\mathfrak{h}$.
- Given a manifold N , let $\text{Emb}(N, M)$ be the space of smooth embeddings $N \hookrightarrow M$, equipped with the Whitney C^2 -topology.

As the cut locus is closed, we have the maps

$$\begin{aligned} \text{Cut} : M \times \mathcal{G} &\rightarrow \mathcal{K} \\ (p, \mathfrak{g}) &\mapsto \text{Cut}(p, \mathfrak{g}), \end{aligned}$$

$$\begin{aligned} \text{Cut} : \text{Emb}(N, M) \times \mathcal{G} &\rightarrow \mathcal{K} \\ (f, \mathfrak{g}) &\mapsto \text{Cut}(f(N), \mathfrak{g}), \end{aligned}$$

We also have the map

$$\begin{aligned} \text{Inj} : \text{Emb}(N, M) \times \mathcal{G} &\rightarrow (0, \infty) \\ (f, \mathfrak{g}) &\mapsto \text{Inj}(f(N), \mathfrak{g}). \end{aligned}$$

Stability of Cut Locus and Injectivity Radius

Theorem : ([Itoh, Prasad 2026][†])

The cut locus map $\text{Cut} : M \times \mathcal{G} \rightarrow \mathcal{K}$ is continuous.

Recently, we have generalized the result of Sakai [1983].

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Given a closed manifold N , the injectivity radius map

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Proof Sketch of Stability: Ingredients

We have the characterization of convergence in the Hausdorff metric.

Lemma

Suppose M is compact, $X_i, X \subset M$ are closed (and hence compact) sets. Then, $\lim_i d_H(X_i, X) = 0$ if and only if the following holds:

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We also have a useful characterization of a separating point. Given $\Omega \subset M$, consider the cut locus (resp. separating set) of the boundary $\partial\Omega$ inside Ω :

$$\text{Cut}^{\text{in}}(\Omega, g) := \text{Cut}(\partial\Omega, g) \cap \Omega, \quad \text{Sep}^{\text{in}}(\Omega, g) := \text{Sep}(\partial\Omega, g) \cap \Omega.$$

The distance function $u := \text{dist}_g(\partial\Omega, \cdot)$ is *not* C^1 on points of $\text{Sep}^{\text{in}}(\Omega, g)$, and the C^1 singular support of u inside Ω is precisely $\text{Cut}^{\text{in}}(\Omega, g)$.

Lemma

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Lemma

Suppose $g_i \rightarrow g_0$ is a C^2 -convergent sequence of metrics. Given $x \in \text{Cut}(N, g_0)$, there exists a sequence $x_i \in \text{Cut}(N, g_i)$ such that $\lim_i x_i = x$.

Proof : Since $\text{Cut}(N, g_0) = \overline{\text{Sep}(N, g_0)}$, assume $x \in \text{Sep}(N, g_0)$. As $\lim_i \text{Inj}(N, g_i) = \text{Inj}(N, g_0) > 0$, we have $\delta > 0$ such that $\delta < \text{Inj}(N, g_i)$. Fix the sets and the distance functions:

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As $g_i \xrightarrow{C^2} g_0$, we have $u_i \rightarrow u_0$ uniformly on compacts. As $x \in \text{Sep}(N, g_0) = \text{Sep}^{\text{in}}(N, g_0)$, we have a C^1 function φ near x such that $u - \varphi$ attains a strict local maxima at x , and

$$\|d_x \varphi\|_{g_0} < 1.$$

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$$\lim_i \|d_{x_i} \varphi\|_{g_i} = \|d_x \varphi\|_{g_0} < 1.$$

Hence, $\|d_{x_i} \varphi\|_{g_i} < 1$ for $i \gg 1$. But then $x_i \in \text{Sep}^{\text{in}}(\Omega_i, g_i) = \text{Sep}(N, g_i)$, concluding the proof. □

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$$\|d_x \varphi\|_{\mathbf{g}_0} < 1.$$

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Hence, $\|d_{x_i} \varphi\|_{\mathbf{g}_i} < 1$ for $i \gg 1$. But then $x_i \in \text{Sep}^{\text{in}}(\Omega_i, \mathbf{g}_i) = \text{Sep}(N, \mathbf{g}_i)$, concluding the proof. □

Proof Sketch of Stability

Lemma

Suppose $\mathbf{g}_i \rightarrow \mathbf{g}_0$ is a C^2 -convergent sequence of metrics. Given $x \in \text{Cut}(N, \mathbf{g}_0)$, there exists a sequence $x_i \in \text{Cut}(N, \mathbf{g}_i)$ such that $\lim_i x_i = x$.

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Proof : Since $\text{Cut}(N, \mathbf{g}_0) = \overline{\text{Sep}(N, \mathbf{g}_0)}$, assume $x \in \text{Sep}(N, \mathbf{g}_0)$. As $\lim_i \text{Inj}(N, \mathbf{g}_i) = \text{Inj}(N, \mathbf{g}_0) > 0$, we have $\delta > 0$ such that $\delta < \text{Inj}(N, \mathbf{g}_i)$. Fix the sets and the distance functions:

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Proof : Since $\text{Cut}(N, \mathbf{g}_0) = \overline{\text{Sep}(N, \mathbf{g}_0)}$, assume $x \in \text{Sep}(N, \mathbf{g}_0)$. As $\lim_i \text{Inj}(N, \mathbf{g}_i) = \text{Inj}(N, \mathbf{g}_0) > 0$, we have $\delta > 0$ such that $\delta < \text{Inj}(N, \mathbf{g}_i)$. Fix the sets and the distance functions:

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Suppose $g_i \rightarrow g_0$ is a C^2 -convergent sequence of metrics. Given $x_{i_j} \in \text{Cut}(N, g_{i_j})$ with $x = \lim_j x_{i_j}$, we have $x \in \text{Cut}(N, g_0)$.

Proof : Without loss of generality, assume $x = \lim_i x_i$, with $x_i \in \text{Sep}(N, g_i)$. If $x \notin \text{Cut}(N, g_0) = \text{Cut}^{\text{in}}(\partial\Omega_0, g_0)$, then in a neighborhood there are *unique* g_0 -distance-minimizers from $\partial\Omega_0$. Using $\lim_i \text{Inj}(N, g_i) = \text{Inj}(N, g_0)$, and $g_i \xrightarrow{C^2} g_0$, we derive that in a neighborhood of x there are *unique* g_i -distance-minimizers from $\partial\Omega_i$. This implies $x_i \notin \text{Sep}(N, g_i)$, a contradiction. $\square$

Proof of the Continuity : The proof of the continuity of the cut locus map now follows from the previous two lemmas, and the characterization of the convergence in the Hausdorff metric. $\square$

Corollary

Suppose $g_i \rightarrow g_0$ is a C^2 -convergent sequence of metrics, and $f_i \rightarrow f_0$ is a C^2 -convergent sequence of embeddings $N \hookrightarrow M$; set $N_i = f_i(N)$. Let $\mathbf{n}_i \in TN_i^{\perp_{g_i}}$ be unit normal vectors converging to some $\mathbf{n}_0$, necessarily in $TN_0^{\perp_{g_0}}$ with $\|\mathbf{n}_0\|_{g_0} = 1$. Then, $\lim_i \rho_{g_i}(N_i, \mathbf{n}_i) = \rho_{g_0}(N_0, \mathbf{n}_0)$.

The above strengthens the classical result about the continuity of the cut time map of a fixed submanifold, with respect to a fixed metric.

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Focal Points

A cut point may not always be a separating point.

Let N be an ellipse in $\mathbb{R}^2$, with focal points e, e' .

$\text{Sep}(N) = \text{Open interval between } e \text{ and } e'.$

For some x on the open interval, we have $[p_1, x]$ and $[p_2, x]$ as N -segments.

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Any extension of the unique N -segment $[a, e]$ (or $[b, e']$) fails to be distance minimizing from N .

Definition

Given $N \hookrightarrow M$, a point $q \in M$ is called a **focal point** of N if it is a *critical value* of the normal exponential map $\exp^\perp : TN^\perp \rightarrow M$

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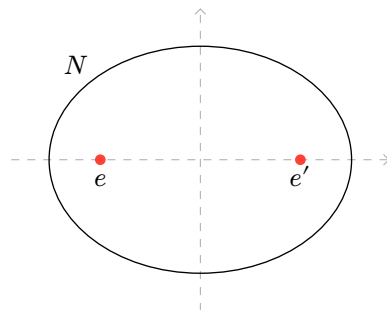
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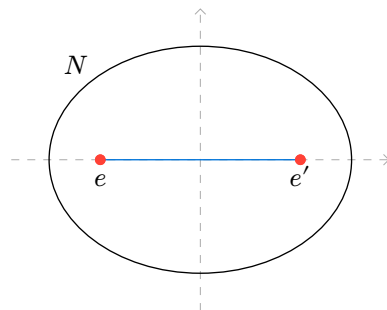
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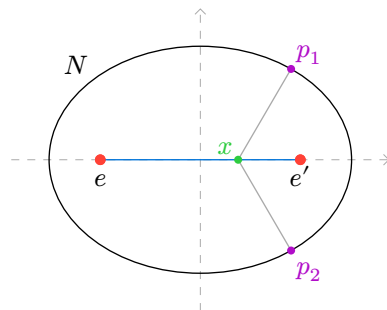
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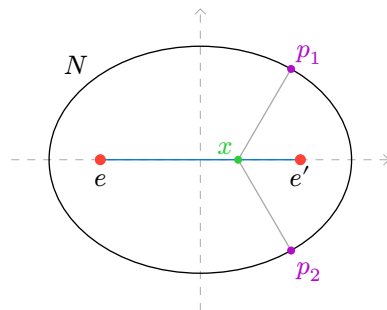
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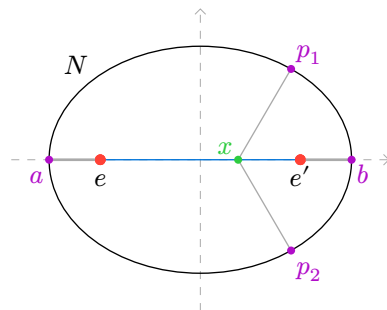
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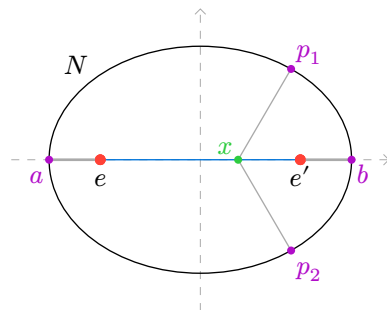
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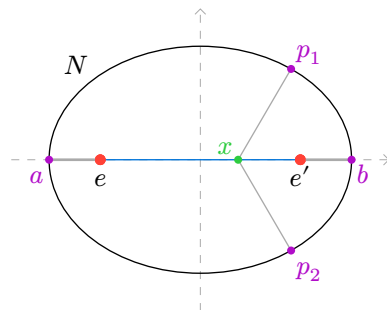
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Intersection of Cut and Focal Loci

In 1959, Rauch made the following conjecture.

Conjecture : (Rauch [1959][†])

For any point p in a compact, simply connected, Riemannian manifold (M, g) , the cut locus $\text{Cut}(p, g)$ must have a conjugate point of p along some distance-minimizing geodesic from p .

The conjecture is true for S^2 , and for symmetric spaces.

Theorem : (Weinstein [1968][‡])

If M is a compact manifold without boundary, not homeomorphic to S^2 , then there exists a metric g on M and a point $p \in M$, such that the cut locus $\text{Cut}(p, g)$ has no conjugate point of p along any distance-minimizing geodesic from p .

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Suppose M is a compact, connected, manifold without boundary, with $\dim M \geq 3$. Let $N \subset M$ be a closed, embedded submanifold, each component being without boundary, and having codimension ≥ 1 . Then, there exists a metric g on M such that the cut locus $\text{Cut}(N, g)$ contains no focal point of N along an N -segment.

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Intersection of Cut and Focal Loci

In 1959, Rauch made the following conjecture.

Conjecture : (Rauch [1959][†])

For any point p in a compact, simply connected, Riemannian manifold (M, g) , the cut locus $\text{Cut}(p, g)$ must have a conjugate point of p along some distance-minimizing geodesic from p .

The conjecture is true for S^2 , and for symmetric spaces.

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An Estimate on the First Focal Time

Principal curvature : generalization of the curvature of a planar curve.

Sectional curvature : generalization of the Gaussian curvature of a surface.

Theorem : (Warner [1966][†])

Let $H \subset M$ be a codimension 1 submanifold in a compact Riemannian manifold (M, g) , and $\mathbf{n}$ be a unit vector field normal to H at every point. Suppose $\Delta_H, \kappa_M > 0$ are real numbers such that

- the principal curvatures of H along $\mathbf{n}$ are $> -\Delta_H$, and
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Set

$$\varepsilon_H := \frac{1}{\sqrt{\kappa_M}} \arctan \left(\frac{\sqrt{\kappa_M}}{\Delta_H} \right).$$

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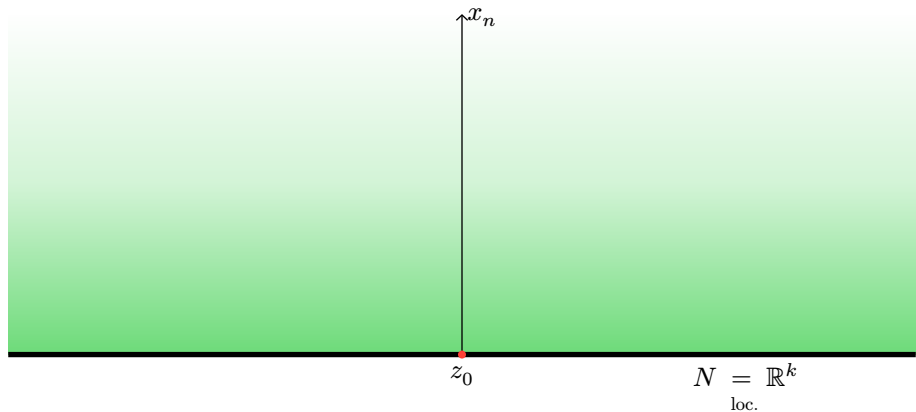
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Step 1: Choosing an Initial Metric and a Tubular Neighborhood

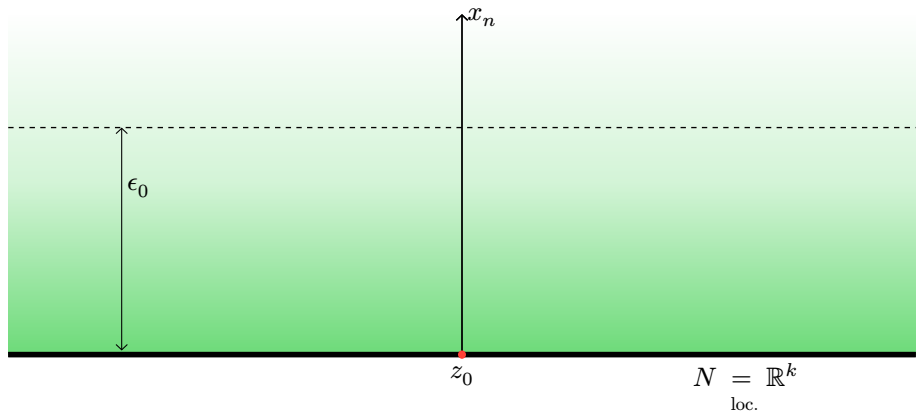
For $z_0 \in N$, get a metric which is Euclidean near z_0 and N is isometric to $\mathbb{R}^k \times \{0\} \subset \mathbb{R}^n$. Choose an ϵ_0 -tubular neighborhood of N , which is $\mathbb{R}^k \times \mathbb{D}_{\epsilon_0}^{n-k}$ near z_0 . Get a smaller tubular neighborhood Ω , whose boundary P is flat and parallel to $\mathbb{R}^{n-1} \times \{0\}$ near z_0 .



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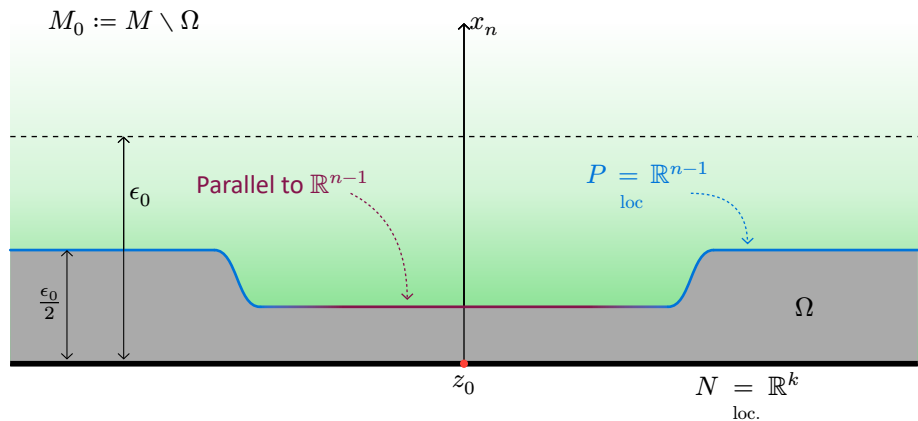
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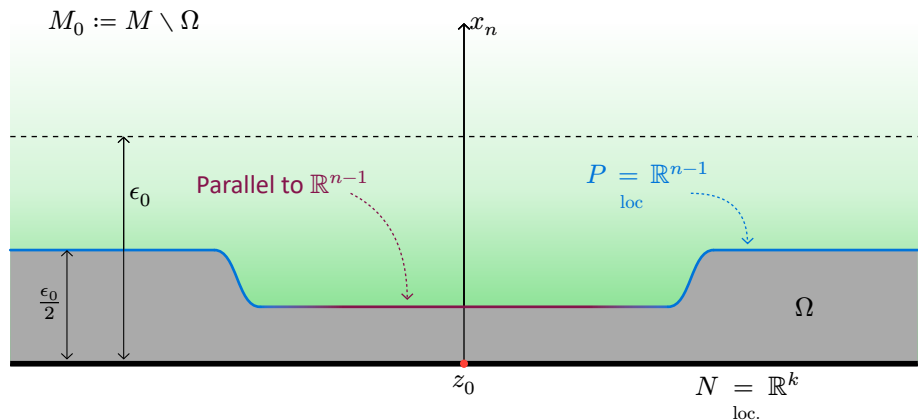
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Theorem : (Prasad, Schick)

There exists $\Delta_{\text{curve}} > 0$, such that we can get a regular curve in $M_0 \setminus P$, with geodesic curvature strictly bounded by Δ_{curve} , which is arbitrarily dense in a given compact set $\mathcal{K} \subset M_0 \setminus P$. The curve can be assumed to be embedded if $\dim M \geq 3$.

Fix $\Delta_P, \kappa_M > 0$, such that principal curvatures of P are $> -\Delta_P$ along the outward normal directions, and sectional curvatures of M are $< \kappa_M$. Set,

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Get ε_0 -neighborhood of P , and set $\mathcal{K}$ as the region away from N , with boundary Q .

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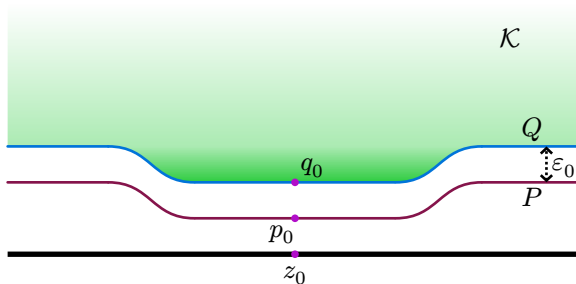
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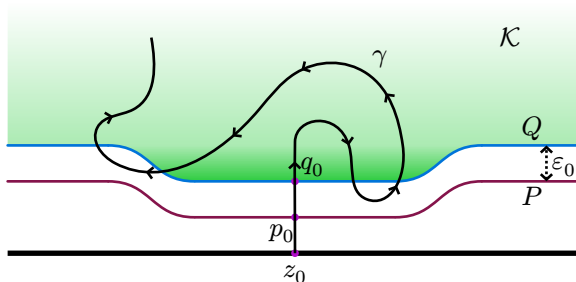
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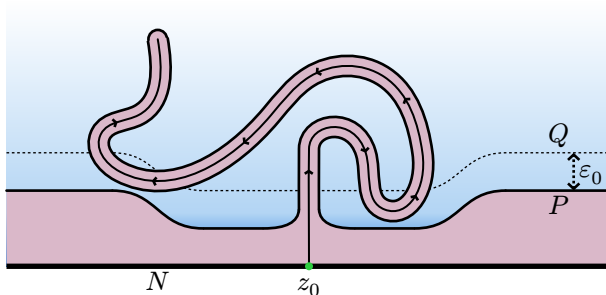
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Step 3: Constructing a New Tubular Neighborhood

Next, we get a tube around the curve γ , and join the boundary smoothly to P .



The new region $\tilde{\Omega}$ is a tubular neighborhood of N , with boundary $\mathcal{B}$. We estimate the principal curvatures of $\mathcal{B}$.

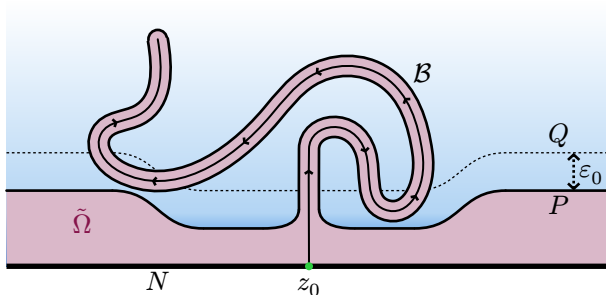
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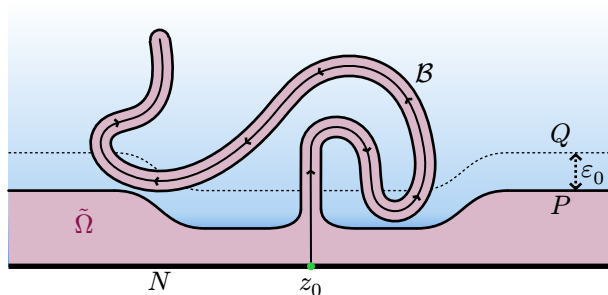
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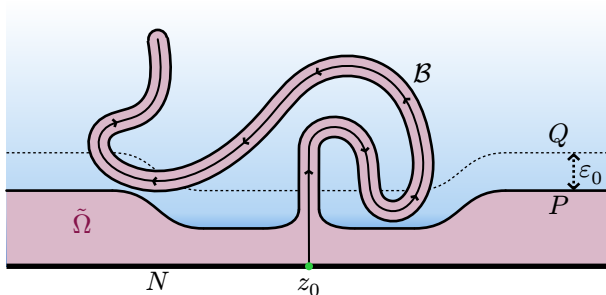
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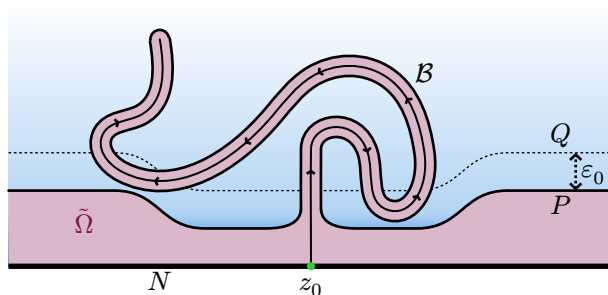
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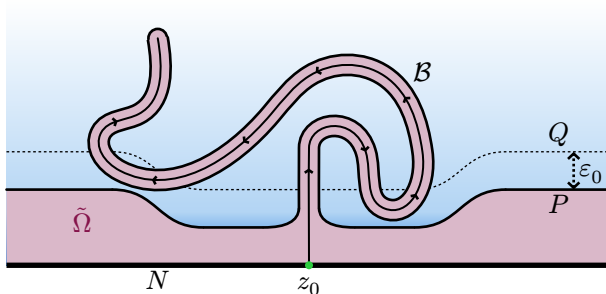
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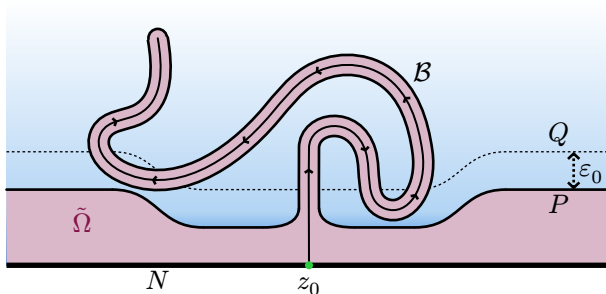
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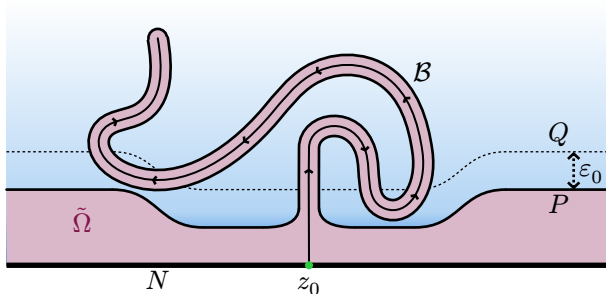
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Hence, by the $\frac{\epsilon_0}{2}$ -denseness, a cut point of $\mathcal{B}$ on the outside $M_0 \setminus \tilde{\Omega}$ is *not* a focal point of $\mathcal{B}$ along any $\mathcal{B}$ -segment.

Step 3: Constructing a New Tubular Neighborhood

Next, we get a tube around the curve γ , and join the boundary smoothly to P .



The new region $\tilde{\Omega}$ is a tubular neighborhood of N , with boundary $\mathcal{B}$. We estimate the principal curvatures of $\mathcal{B}$.

- By an argument of Weinstein, around γ , the principal curvatures of $\mathcal{B}$ along the outward normal are $> -\sqrt{\frac{32}{27}}\Delta_{\text{curve}} \geq -\Delta_{\text{final}}$.
- On P , the bound is $> -\Delta_P \geq -\Delta_{\text{final}}$.
- The funnel-shaped join is in a Euclidean region, where we can perform explicit computations.

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Step 4: Changing the Metric inside the Tubular Neighborhood

Theorem : (, Prasad, Schick)

There exists a new metric g on M such that the following holds.

- g is unchanged near $\mathcal{B}$, the boundary of the tubular neighborhood $\tilde{\Omega}$.
- g is unchanged on the complement of $\tilde{\Omega}$.
- $\tilde{\Omega}$ is a *unit normal neighborhood* of N with respect to N .

Proof of Main Theorem : With the new metric g ,

- every $\mathcal{B}$ -segment $\gamma : [0, \epsilon] \rightarrow M_0 \setminus \tilde{\Omega}$ extends back to an N -segment $\tilde{\gamma} : [-1, \epsilon] \rightarrow M$, and
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- cut points of N are precisely the cut points of $\mathcal{B}$ in $M_0 \setminus \tilde{\Omega}$, and
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Some Consequences

Theorem : ([Prasad, Schick])

Let $N_1, N_2 \subset M$ be compact, connected, disjointly embedded, submanifolds, without boundary. Then, the following are equivalent.

- There is a metric g on M such that $N_2 = \text{Cut}(N_1, g)$.
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Given a closed manifold M , if there exists a function $f : M \rightarrow \mathbb{R}$ with exactly two critical values a, b , and if $S^p = f^{-1}(a)$, $S^q = f^{-1}(b)$, with $p \neq q$, then M is homeomorphic to S^{p+q+1} .

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Let (M, h) be a closed Riemannian manifold. Suppose $N \subset M$ is a closed submanifold such that $\text{Cut}(N, h)$ is free of focal points of N along N -segments. Then, there exists a metric g on M such that $\text{Cut}(N, g)$ is a finite subcomplex of M (and still free of focal points).

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[‡] Antonio Lerario, Chiara Meroni, and Daniele Zuddas. *On smooth functions with two critical values*. *Rev. Mat. Complut.* (2024).

[§] Aritra Bhowmick. *Stability of the Injectivity Radius and the Cut Locus of Submanifolds under Perturbations*. *Comm. Anal. Geom.* (to appear) (2026).

Fibrations with a Section

Given a fibration $F \hookrightarrow E \rightarrow B$ with a (homotopy) section $s : B \rightarrow E$, there is a natural map

$$\theta : \Omega B \times \Omega F \rightarrow \Omega E,$$

where ΩX denotes the loop space of a (based) space X . Classically, it is known that θ is a homotopy equivalence. Loop spaces are homotopy associative topological monoids, i.e., H -spaces.

Theorem : (Basu, __, Samanta [2025][†])

θ induces a Lie algebra map in the homotopy groups (where the Lie brackets are given by Whitehead brackets) if and only if the James brace products $\{, \}_s : \pi_i B \times \pi_j F \rightarrow \pi_{i+j-1} F$ vanish identically.

θ is an H -map if and only if the *generalized* James brace product map vanishes identically.

In the paradigm of Stasheff, H -spaces are A_2 -spaces, and a loop space is actually an A_∞ -space. A natural question is :

when is θ an A_n -map, for $2 \leq n \leq \infty$?

Initial investigations indicate connections with polytopes known as resultohedra, their higher homotopy associative generalizations, and with the power and product maps of monoids.

[†] Somnath Basu, Aritra Bhowmick and Sandip Samanta. *On the James brace product: Generalization, relation to H -splitting of loop space fibrations & the J -homomorphism*
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Thank you!